

8. (a) Keeping warm / 88 W

1A 1

$$(b) \quad R_1 = \frac{V^2}{P} = \frac{220^2}{88} \\ = 550 \, \Omega$$

1M

1A 2

$$(c) \quad \text{Total current } I_0 = \frac{P_0}{V} = \frac{550}{220} = 2.5 \, \text{A} \\ \text{Current in } R_1, I_1 = \frac{220}{550} = 0.4 \, \text{A} \\ \text{Current in } R_2, I_2 = 2.5 - 0.4 = 2.1 \, \text{A}$$

|    |                                                                       |
|----|-----------------------------------------------------------------------|
| Or | Power to $R_2$ , $550 \, \text{W} - 88 \, \text{W} = 462 \, \text{W}$ |
|    | Current in $R_2$ , $I_2 = \frac{P_2}{V} = \frac{462}{220}$            |
|    | $= 2.1 \, \text{A}$                                                   |

1M

1M

1A 3

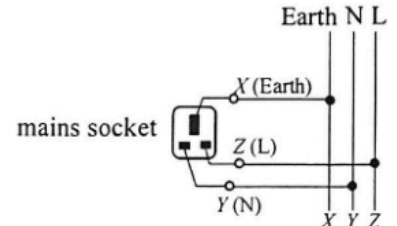
$$(d) \quad \text{Peak current} = \sqrt{2} (2.5 \, \text{A}) \\ = 3.54 \, \text{A}$$

1M

1A 2

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------|
| <p>8. (a)</p> $P = \frac{V^2}{R}$ $500 = \frac{220^2}{R}$ $R = 96.8 \, \Omega$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1A                         | Accept 97 $\Omega$           |
| <p>(b) As the total resistance of the circuit for mode X is doubled,</p> $\text{total power dissipation} = \frac{V^2}{2R}$ $= \frac{220^2}{2 \times 96.8} = 250 \, \text{W}$                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1M                         |                              |
| <p><i>Alternative method:</i><br/>As the voltage across each heating element (1 and 2) is halved, power dissipated in each of the heating element</p> $P_1 \text{ or } P_2 = \frac{500}{4} = 125 \, \text{W} \text{ (since } P \propto V^2 \text{)}$ <p>Or <math>P_1 \text{ or } P_2 = \frac{V^2}{R} = \frac{110^2}{96.8} = 125 \, \text{W}</math></p> <p>Or <math>i = \frac{V}{R_1 + R_2} = \frac{220}{2 \times 96.8} = 1.14 \, \text{A}</math></p> $P_1 \text{ or } P_2 = i^2 R = 1.14^2 \times 96.8 = 125 \, \text{W}$ <p>Total power dissipation = <math>2 \times 125 \, \text{W} = 250 \, \text{W}</math></p> | 1A<br>1M<br>1M<br>1M<br>1A | Accept 249 W for 97 $\Omega$ |
| <p>(c) In mode Z, the equivalent resistance of the heating elements is the least as they are connected in parallel, hence, under the same voltage, the total power dissipation is the largest since <math>P = \frac{V^2}{R}</math>.</p>                                                                                                                                                                                                                                                                                                                                                                            | 2<br>1A<br>1A              |                              |
| <p>(d) (i) For mode Z,<br/>the total power dissipated = <math>500 + 500 = 1000 \, \text{W}</math></p> $I_z = \frac{P}{V} = \frac{1000}{220} = 4.55 \, \text{A}$ <p><i>Alternative method:</i></p> $R_{\text{eq}} = \frac{96.8}{2} \, \Omega = 48.4 \, \Omega$ $I = \frac{220 \, \text{V}}{48.4 \, \Omega} = 4.55 \, \text{A}$ <p>Most suitable value of fuse = 5 A</p>                                                                                                                                                                                                                                             | 2<br>1M+1M<br>1M+1M<br>1A  |                              |
| <p>(ii) Although the heater still works in either connection, it is dangerous for switch S to be fitted in wire B (neutral) as the heater / cable would still be live even when the switch was turned off.</p>                                                                                                                                                                                                                                                                                                                                                                                                     | 3<br>1A<br>1A              |                              |
| <p>(iii) Wire C (Earth).<br/>Current would be conducted from the case through this wire to the Earth.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2<br>1A<br>1A              | 1+1                          |

|                                                                                                                                                                                                                                                                                               |                |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--|
| 8. (a) (i) To terminals X                                                                                                                                                                                                                                                                     | 1A             |  |
|                                                                                                                                                                                                                                                                                               | 1              |  |
| (ii) $P = IV$<br>$800 \text{ W} = I(220 \text{ V})$<br>$I = 3.636364 \text{ A}$<br>$\approx 3.64 \text{ A}$                                                                                                                                                                                   | 1M<br>1A       |  |
|                                                                                                                                                                                                                                                                                               | 2              |  |
| (iii) $800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$<br>$P_{\text{keep warm}} = \frac{V^2}{4R}$<br>$= 800\left(\frac{1}{5}\right) = 160 \text{ W}$                                                                                                                                  | 1M<br>1M<br>1A |  |
| <u>Alternative (I)</u><br>$800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$<br>$R = 75.625 \Omega$<br>$P_{\text{keep warm}} = \frac{V^2}{4R}$<br>$= \frac{220^2}{4(75.625)}$<br>$= 160 \text{ W}$                                                                                      | 1M<br>1M<br>1A |  |
| <u>Alternative (II)</u><br>Power $\propto \frac{1}{\text{resistance}}$<br>$\frac{P_{\text{keep warm}}}{P_{\text{heating}}} = \frac{R_{\text{eq}}}{4R}$<br>$P_{\text{keep warm}} = P_{\text{heating}} \left( \left( \frac{1}{R} + \frac{1}{4R} \right)^{-1} / 4R \right)$<br>$= 160 \text{ W}$ | 1M<br>1M<br>1A |  |
|                                                                                                                                                                                                                                                                                               | 3              |  |
| (b) (i) Electrical energy (consumed)                                                                                                                                                                                                                                                          | 1A             |  |
|                                                                                                                                                                                                                                                                                               | 1              |  |
| (ii) (1) only the fuse blows                                                                                                                                                                                                                                                                  | 1A             |  |
| (2) only the RCCB cuts off the supply                                                                                                                                                                                                                                                         | 1A             |  |
|                                                                                                                                                                                                                                                                                               | 2              |  |

|                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                          |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--|
| <p>8. (a)</p>                                                                                                                                                                                                                                                                                                           | 1A                                                                       |  |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                                        |  |
| <p>(b) (i)</p> <ul style="list-style-type: none"> <li>- If one of the lighting sets / circuits fails, the other (in parallel) can still operate, i.e. both work independently.</li> <li>- Both can work at the rated power.</li> <li>- Any reasonable answer</li> </ul> <p>(ii)</p> $P = IV$ $(300 + 450) = I(220)$ $I = 3.409091 \text{ A} \approx 3.41 \text{ A}$ <p>Thus 5 A fuse should be used.</p> | <p>1A</p> <p>Any ONE</p> <p>1</p> <p>1M</p> <p>1A</p> <p>1A</p> <p>3</p> |  |
| <p>(c) Electrical energy used per day</p> $= 0.500 \text{ kW} \times 8 \text{ h} + 2 \text{ kW} \times 0.5 \text{ h} + 3 \text{ kW} \times 2 \text{ h}$ $= 11 \text{ kW h}$ <p>Cost = \$0.9 / kW h <math>\times</math> 11 kW h</p> $= \$9.9$                                                                                                                                                             | <p>1M</p> <p>1M</p> <p>1A</p> <p>3</p>                                   |  |

8. (a) Because  $L_1$  and  $L_2$  (connected in series) are shorted by BCDE

1A

Accept p.d. across each low-beam headlight ( $L_1$  or  $L_2$ ) is 0.

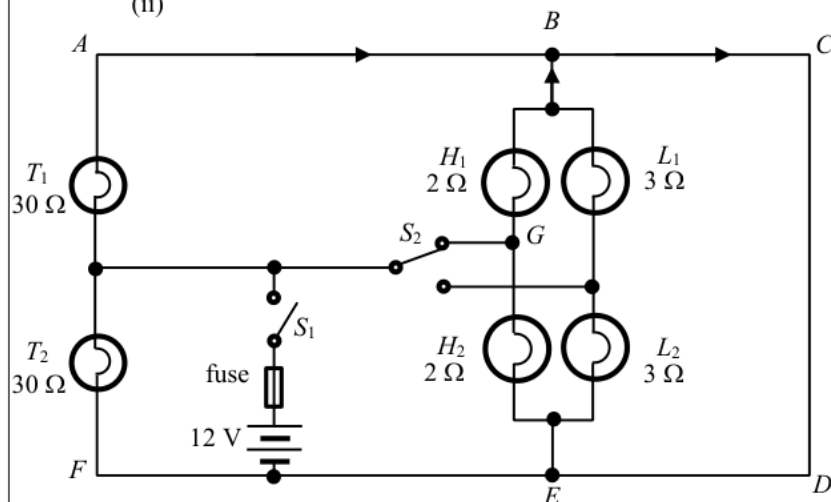
1

(b) (i) p.d. across  $T_2 = 12$  V

1A

1

(ii)



largest current in branch BC

1A

1A

Any two currents correctly marked.  
All currents correct.

1A

3

(c) Power supplied  $P = 2 \times \left( \frac{12^2}{30} + \frac{12^2}{2} \right)$   
 $= 153.6 \text{ W} = 154 \text{ W}$  (3 sig. fig.)

1M

1A

1M for same voltage across each bulb ( $T_1$ ,  $T_2$ ,  $H_1$  and  $H_2$ )  
e.c.f. from (b)(i)

$$\frac{V^2}{R_{eq}} = \frac{12^2}{R_{eq}} = 153.6 \quad \left( \text{or } \frac{1}{R_{eq}} = \frac{1}{30} + \frac{1}{30} + \frac{1}{2} + \frac{1}{2} \right)$$

1M

1A

1M for all bulbs ( $T_1$ ,  $T_2$ ,  $H_1$  and  $H_2$ ) in parallel

$$R_{eq} = 0.9375 \Omega \approx 0.938 \Omega \approx 1 \Omega \quad (3 \text{ sig. fig.})$$

4

(d) max. current  $I_{\max} = \frac{V}{R_{eq}} = \frac{12}{0.9375} = 12.8 \text{ A}$

1M

e.c.f. from (c)

$$\left( \text{or } I_{\max} = \frac{P}{V} = \frac{153.6}{12} = 12.8 \text{ A} \right)$$

1A

1A for 12.8 A and correct conclusion

Since the rating is slightly higher than the max. current (when high-beam headlights and taillights are switched on), it is suitable.

2